FINAL SPIRIT - EXAM SAVER @ TOP 50

Inorganic Chemistry

1. Write the salient features/postulates of crystal field theory and its limitations.
2. Explain the crystal field splitting of d-orbitals in octahedral compounds by taking one example.
3. Discuss the Gouy's method for the determination of magnetic susceptibility.
4. Explain Pearson's classification of hard and soft acid Bases (HSAB). Explain the Pearson's concept of hardness and softness.
5. Explain the determination of composition of complexes using Job's method.
6. Explain the determination of composition of complexes using Mole ratio method
7. Write the structure and functions of Chlorophyll.
8. Write the structure and functions of Haemoglobin.
9. Explain Z- scheme.
10. What is Chelate Effect? Explain in detail using thermodynamic parameters.
11. Calculate the magnetic moments of the following :-

    (i) $[CoF_6]^{-3}$ (ii) $[CoCl_4]^{-2}$ (iii) $[Fe(CN)_6]^{-3}$ (iv) $[Ni(CN)_4]^{-2}$ (v) $[Mn(H_2O)_6]^{+2}$

    (vi) $[Ni(CO)_4]$ (vii) $[Co(NH_3)_6]^{+3}$ (viii) $[VCl_4]^{-}$
12. What is Zieglar Natta catalyst and its applications.
13. Explain the crystal field splitting of d-orbitals in tetrahedral compounds.
14. Write the Biological significance of Na^+, K^+, Mg^{+2}, Ca^{+2}, Cl^-, Fe^{+2}, Co^{+2}, Ni^{+2}, Cu^{+2}, Zn^{+2} (ionic forms)

Organic chemistry

15. Define the following :
    a) Mutarotation
    b) Epimers with Example
    c) Epimerisation
    d) Anomers with Example
    e) Zwitter ion and Lactams
    f) Isoelectric point with Example.
16. Structural Elucidation of glucose (or) Evidence of pentahydroxy, straight chain, Aldehyde group in open chain.

(17) Give the evidence for cyclic structure of glucose.

(18) Discuss Osazone formation from glucose.

(19) Explain the following interconversions with equations

(i) D-Glucose → D-Arabinose

(ii) D-Glucose → D-Fructose

(iii) D-Fructose → D-Glucose

(20) Explain the preparation of Amino acids by (i) Strecker's Synthesis
(ii) Malonic Ester Synthesis

(21) Write the classification of Amino acids based on the nature and structure.

(22) Explain the aromaticity and reactivity of pyrrole, furan and thiophene.

(23) Explain the acidic nature of pyrrole and basic nature of pyridine.

(24) Write Paal Knorr Synthesis of pyrrole, Furan and thiophene and Explain chichibabin reaction.

(25) Explain (i) Diels-Alder reaction
(ii) Sulphonation of thiophene.

Physical chemistry

(26) Define: a) Rate of Reaction
b) Rate equation / Rate Law
c) Rate constant and units
d) order of reaction.

(27) Factors affecting rate of reaction

(28) Difference b/w order and Molecularity

(29) What is Pseudo First order reaction with Examples.

(30) Define First order reaction. Derive integral form of First order rate constant? Half life of First order and units.

(31) Define Second order reaction. Derive integral form of Second order rate constant? Half life of second order and units.

(32) The half life of 1st order reaction was found to be 10min at a certain temperature. What is rate constant.

(33) Difference b/w a) photochemical and thermal reactions
b) Fluorescence and phosphorescence.

(34) Explain Jablonski diagram and explain different photophysical processes on it.

(35) Explain Grothus- Drapper's law and Stark Einstein's law (or) Laws of photochemistry.

(36) Define Quantum yield and Explain its types.

General Chemistry :-

(37) Define conductors, semiconductors and Insulators with Example.

(38) Explain Valence Bond Theory in metals and their limitations.

(39) Describe in detail about Band Theory in metals.

(40) Explain the classification of Semi-conductors.

(41) Explain about the formation of P-type and N-type Semiconductors.

(42) Explain Mannich reaction and mechanism.

(43) Explain Knoevengel condensation

(44) Synthetic applications of acetoacetic ester.

(45) What is Malonic Ester & Write 2 synthetic applications.

(46) What is Tyndall effect.

(47) What are colloids and its classification.

(48) Explain Gold number.

(49) Write Freundlich adsorption isotherm

(50) Discuss the Langmuir theory of adsorption.